



CHAPTER

Feed and Process Pumps

Feed, diluate and concentrate pumps on each SonixED skid

IN THIS CHAPTER

9.1	Overview and function	66
9.2	Key specifications	67
9.3	Operation	68
9.4	Maintenance	69
9.5	Troubleshooting	70

9.1 Overview and function

Each SonixED™ skid carries three process pumps — feed, diluate and concentrate — that circulate water through the stack(s) at the required flow and pressure. Pumps are magnetic-driven (sealless) and inverter-(VFD-)controlled so flow can be adjusted without throttling valves.

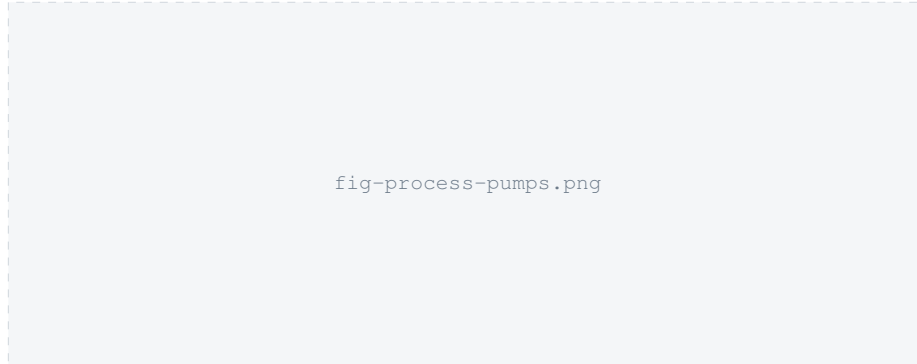


Figure 9.1. Suggested figure: photograph of the three process pumps on a skid, labeled feed/diluate/concentrate, with their VFD units visible.

9.2 Key specifications

- Always 3 pumps per skid (feed, diluate, concentrate).
- Magnetic-driven, inverter-driven, polypropylene/stainless-steel construction, seawater-compatible.
- Pretreatment feed pump (pilot): Grundfos CRN 10-5 A-FGJ-A-E-HQQE, flanged connections.
- Rated hydraulic flow per stack: 24 m³/day (SonixED-S datasheet).
- Each pump is driven by its own VFD (control tags VFD_1_D1...VFD_6_E2 in the pilot configuration).

9.3 Operation

Pump speed is set by the PLC via the VFDs to hold the process at its target flow/pressure. Manual speed override, if available, is reached via the `Manual` mode described in chapter 6.

9.4 Maintenance

TODO: no pump-specific maintenance schedule (bearing/seal inspection, impeller check interval) was found. Magnetic-driven pumps have no shaft seal to replace, which reduces routine maintenance versus a conventional centrifugal pump, but confirm this against the actual pump model fitted.

9.5 Troubleshooting

Each pump's VFD reports a `FaultWord` that should be checked first on any pump-related alarm. Typical causes to check, in order:

- **Pump not running / VFD fault:** check `FaultWord` on the relevant VFD tag; check incoming supply and control-voltage presence (chapter 14).
- **Low flow / high pressure:** check for blockage upstream (chapter 8) or downstream (stack fouling, chapter 10) before assuming a pump fault.
- **Pump running but no output:** check for air lock (magnetic- driven pumps are not self-priming) — TODO confirm priming procedure.